

CAK2HAB3 - Dasar Kecerdasan Artifisial

Reasoning: Introduction to Knowledge-Based System

Lecturer Team





Knowledge and Reasoning



Reasoning

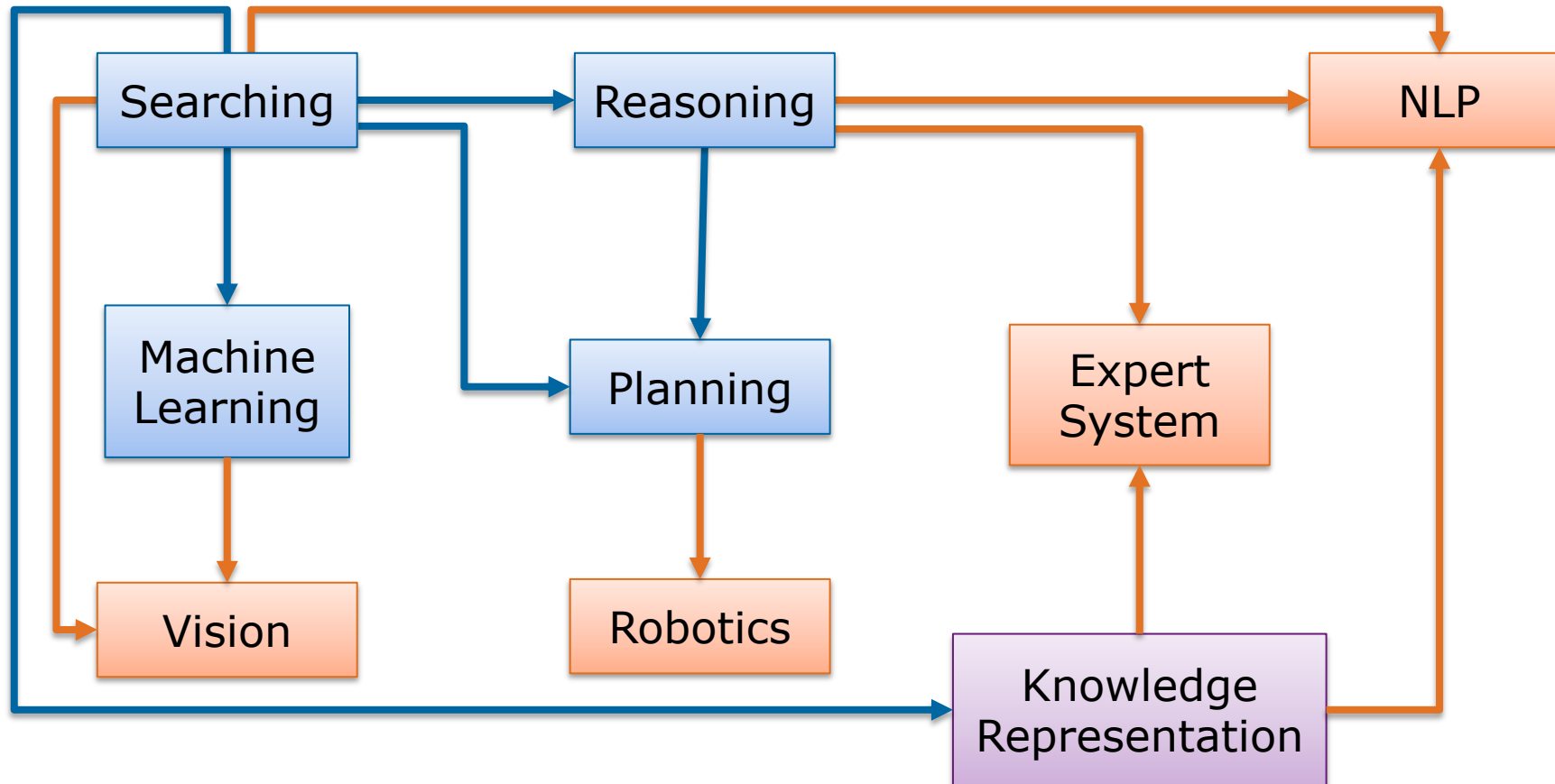
- ▶ System that generates conclusions from available knowledge using logical techniques such as deduction and induction.
- ▶ humans are very good at acquiring new information by combining raw knowledge, experience with reasoning



Knowledge representation and Reasoning (KR)

- ▶ The field of artificial intelligence (AI) dedicated to
REPRESENTING INFORMATION about the world
in a form that a computer system can utilize
to **Solve Complex Tasks**
such as diagnosing a medical condition
or having a dialog in a natural language

Some Areas of AI applications





**How can we Represent
Human Knowledge about The World
and How to do Things?**



The Problem with Natural Language

► Ambiguous

- “I know **more** intelligent people than him.”
- “I saw **the Grand Canyon** flying to New York.”
- “I saw a **747** flying to New York.”
- “I saw **a man on the hill with a telescope**”
- a: He **left the bank** five minutes ago.
b: He **left the bank** five years ago

► We need a formal way of representing the knowledge usually represented in language



Two schools of thought in AI

- ▶ Symbolic:
 - represent knowledge using logic and rules that define inference procedures on logic
- ▶ Statistical:
 - represent knowledge as probabilities based on observations of the world

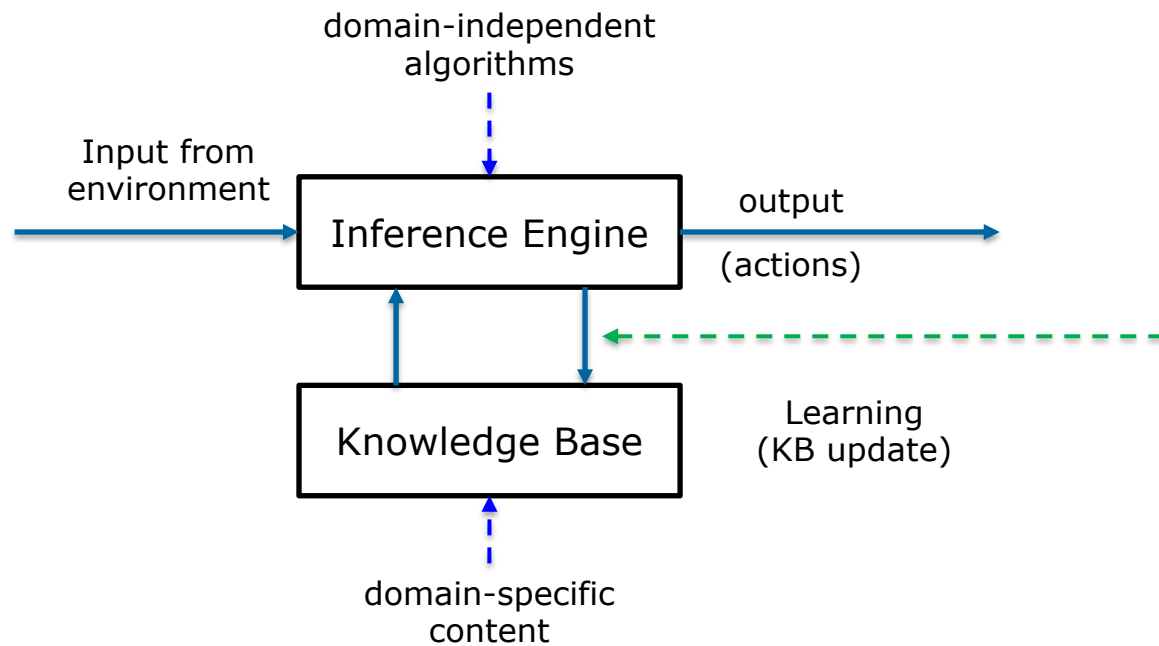


Knowledge-based Logical Agent

- ▶ An agent that **reasons** by using a **knowledge base** and an **inference system** to solve complex problem
 - A knowledge base is a set of representations of facts of the world.
 - Each individual representation is called a sentence
 - The sentences are expressed in a knowledge representation language.



Knowledge-based Logical Agent



Knowledge-based Logical Agent

▶ Knowledge Base :

- set of sentences represented in a knowledge representation language or formal language where it's representing facts about the world

▶ Inference Engine

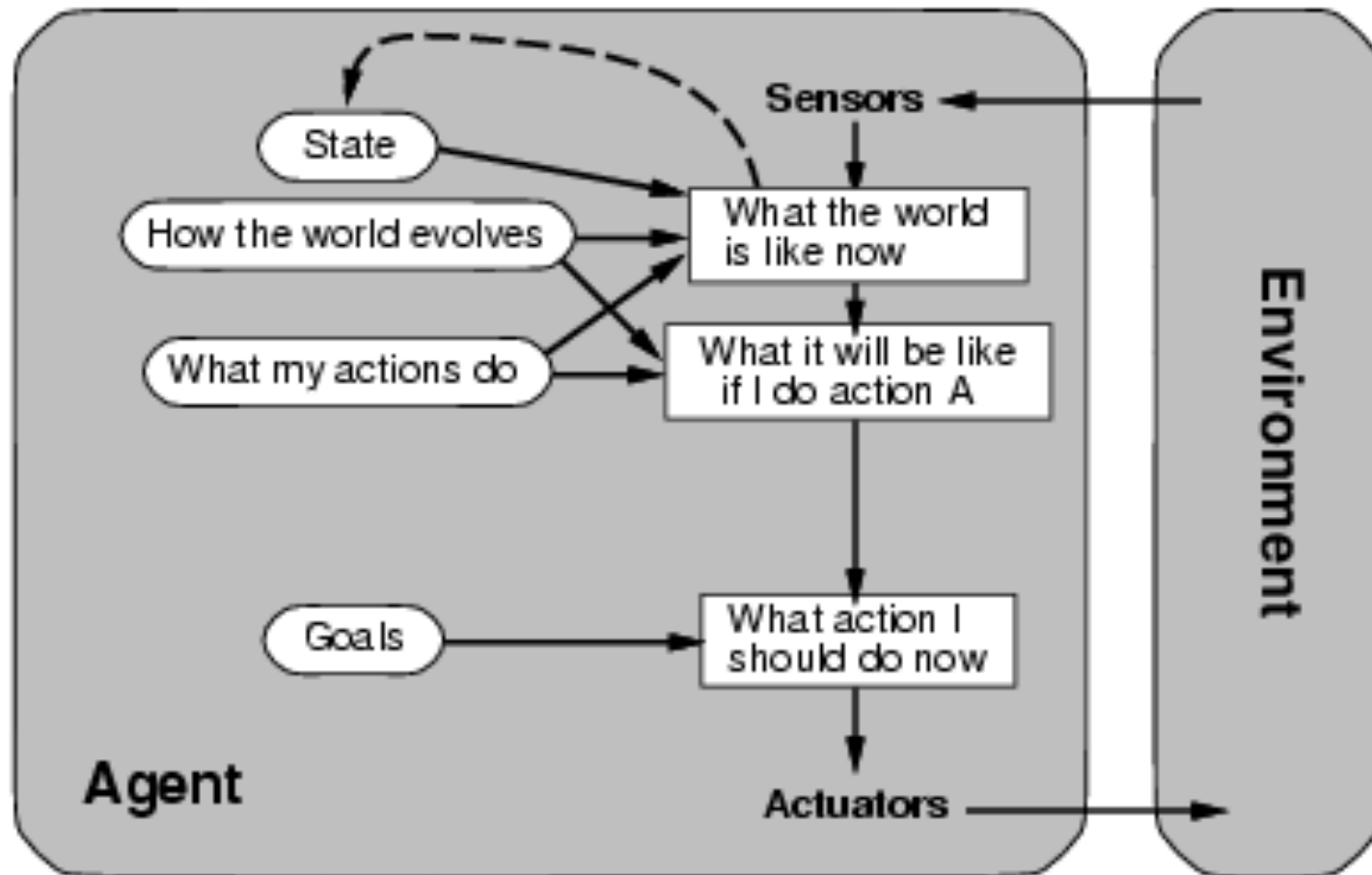
- a component of the system that applies logical rules to the knowledge base to deduce new information
- derives new sentences from the input and KB
- Engine to process the reasoning



Knowledge-based Logical Agent

- ▶ The agent must be able to:
 - Represent states, actions, etc.
 - Incorporate new percepts
 - Update internal representations of the world
 - Deduce hidden properties of the world
 - Deduce appropriate actions

Knowledge-based Logical Agent





Knowledge-based Logical Agent

- The agent operates as follows:
 1. It **TELLs** the knowledge base what it perceives.
 2. It **ASKs** the knowledge base what action it should perform.
 3. It **performs** the chosen action

- Knowledge-based logical agents combine **general knowledge** about the world with current percepts to infer hidden aspects of their state



Knowledge-based Logical Agent

```
function KB-AGENT(percept) returns an action  
  static: KB, a knowledge base  
         t, a counter, initially 0, indicating time  
  
  TELL(KB, MAKE-PERCEPT-SENTENCE(percept, t))  
  action  $\leftarrow$  ASK(KB, MAKE-ACTION-QUERY(t))  
  TELL(KB, MAKE-ACTION-SENTENCE(action, t))  
  t  $\leftarrow$  t + 1  
  return action
```



Five types of Logic

Type of logic	Reference to real world	Type of knowledge known by Agent
Propositional Logic	Facts	True / False / Unknown
First-order Logic	Facts, Object, Relation	True / False / Unknown
Temporal Logic	Facts, Object, Relation, Time	True / False / Unknown
Probability Theory	Facts	Degree of belief [0, 1] (certainty factor)
Fuzzy Logic	Truth values (degree of truth)	Confidence level [0, 1] (certainty factor)



Question?





THANK YOU